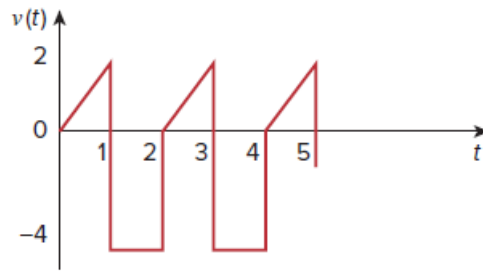


Problem

Find the rms value of the signal shown in Fig. 11.62.

Figure 11.62:



Step-by-step solution

Step 1 of 1

$$V_{rms}^2 = \frac{1}{2} \int_0^2 V(t) dt$$

$$V_{rms}^2 = \frac{1}{2} \left[\int_0^1 (2t)^2 dt + \int_1^2 (-4)^2 dt \right]$$

$$V_{rms}^2 = \frac{1}{2} \left[\frac{4}{3} + 16 \right]$$

$$V_{rms}^2 = 8.6667$$

$$\therefore \boxed{V_{rms} = 2.944 \text{ V}}$$